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CSE 3200: System Development Project [14,Normal, Centered]

AN EFFICIENT IMPLEMENTATION SCHEME FOR MULTIDIMENSIONAL INDEX ARRAY

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John Doe

[14, Normal, Centered] Roll: 2007001

&

Jane Doe

Roll: 2007002

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Department of Computer Science and Engineering
Khulna University of Engineering & Technology
Khulna 9203, Bangladesh

December, 2023

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An Efficient Implementation Scheme for Multidimensional Index Array

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By [*12, Normal, Centered*]

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John Doe

Roll: 2007001

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&

Jane Doe

Roll: 2007002

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Supervisor: [*12, bold, left aligned*]

Mr. Xyz [*12, bold, 0.8" left*]

Professor

Department of …

Khulna Univer…

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_____

Signature

Department of Computer Science and Engineering

Khulna University of Engineering & Technology

Khulna 9203, Bangladesh

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December, 2023

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Acknowledgment

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All the praise to the almighty Allah, whose blessing and mercy succeeded me to complete this project works fairly. I gratefully acknowledge the valuable suggestions, advice and sincere co-operation of …

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Authors

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Abstract

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Write an abstract of your Project within 500 words.

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CHAPTER I

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Introduction

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1.1 Introduction

There are few classes of data structures which are as well understood or as extensively used as arrays. It is quite often for scientific, statistical and engineering applications to have computation on large multidimensional arrays [1,2]. The …

The fast random accessing capability of multidimensional arrays is a fascinating characteristic [3,4] that enables various …

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1.2 Problem Statement

There are many existing array systems to represent multidimensional data such as Traditional …

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1.3 The Realization of Multidimensional Array

There are many data structures already exist to represent multidimensional data. Some of the well-known and prominent data structures are discussed next.

1.3.1 Traditional Multidimensional Array (TMA) [5]

Traditional Multidimensional Array (TMA) is a representation scheme for multidimensional data which represent *n* dimensional data. An illustration of 3dimensional TMA of dimension length 3×4×5 is given in Figure 1.1. F

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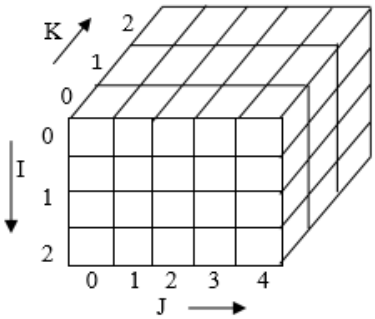

Figure 1.1: A three dimensional TMA with length of dimension 3×4×5.

Table 1.1: Parameters for cost function for TMA

Parameter	Description
n	Number of dimension
d_i	Dimension i, $1 \leq i \leq n$
l_i	Length of dimension d_i
V	Initial volume
s	Number of segments in a subarray

The cost functions are represented as the number of array cells required to access. The parameters that are assumed are described in Table 3.1. All the lengths are in bytes. Some parameters are provided as input while others are derived from input parameters.

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REFERENCES

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